A student performance prediction model using machine learning models in multimodal learning analytics

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Abstract. Multimodal learning analytics in educational data mining is an emerging subject that seeks to enhance student learning outcomes by applying techniques to a wide range of data sources. Recent multimodal learning analysis research has developed models utilizing a range of methodologies. In this study, we investigate various machine learning models utilizing data from multiple sources, ultimately developing a multimodal learning model to predict student performance across four categories: Distinction, Pass, Fail, and Withdraw. The proposed model was experimented on the Open University Learning Analytics Dataset (OULAD) for multi-class and binary classification problems. The multi-class classification results show an Accuracy of 82.31% and an F1-score of 82.82%, while the binary classification results have the highest accuracy of 97.9%.

Keywords: Multimodal Learning Analytics, Data Fusion, Machine learning, Oulad, Oversampling.

1 Introduction

Multimodal learning analysis (MLA) is a method of collecting and integrating data from many different sources, it gives teachers an overview of the learner's learning process [1], [2]. MLA is increasingly popular in recent years in both online and offline learning environments and provides a comprehensive and meaningful way to measure and understand student performance [3]. Data sources can include log data, learner interactions on online platforms, bio-sensors, videos, audio, digital documents and any other source. As a result, integrating or merging data taken from numerous sources/methods is a critical problem for providing a more complete picture of the teaching-learning process [4]. Most previous studies [5], [6], [7], [8], [9] have used the

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OULAD dataset with one or two data sources to test models, and very few studies have used three data sources [6], [10], [11]. In this study, we use different fusion techniques from three different data sources to test the machine learning model. Data fusion techniques have shown higher Accuracy and F1 score compared to using data from a single source. Depending on the particular use case, there are several applications for data fusion techniques in multimodal learning analysis. The data fusion techniques in Fig. 1. may be classified into three types based on when they are used [1]:

• Early fusion: Combines distinct properties of data from several sources into a single vector.

• Late fusion: Use each data source individually, each with its own model, and then combine the outputs of the models.

• Hybrid: A fusion strategy that combines the two methods listed above.

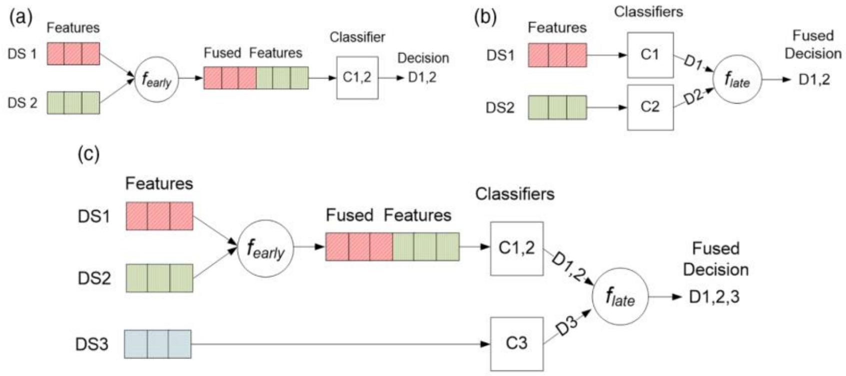

Fig. 1. Different types of MLA data fusion. (a) Early fusion. (b) Late fusion. (c) Hybrid [1].

Multimodal data can be imbalanced in class distribution after fusion. In this study, we use the Oversampling technique to increase the number of samples in minority classes, reducing bias and improving prediction performance. This technique enhances the training data by creating multiple copies of minority class samples, which can be done multiple times and has proven effective [12]. To perform Oversampling, we apply the SMOTE algorithm [13] to retain the original data information, creating new data samples by combining nearby data samples without removing any data points. Additionally, after data fusion, it is necessary to optimize feature selection and reduce the number of features used in the model to reduce computational costs and improve model performance.

Based on the extensive analysis, this study proposes several recommendations to enhance the performance of the student ability prediction model within the context of multimodal learning analytics as follow:

• A student performance classification model based on various data fusion methods from several different sources.
• Use the Oversampling algorithm to balance data.
• Determine the importance of each attribute to improve model effectiveness.

2 Related Work

Several studies have emerged that leverage data from multiple sources to construct models for predicting learner skills [6], [14], [15], [16], [17]. Machine learning and deep learning are the dominant methodological approaches employed in multimodal learning analytics (MLA) research (reference source on MLA methodologies).

Waheed et al. [7] investigated the potential of Artificial Neural Networks (ANN) for predicting at-risk students using multi source data available in the OULAD. Their objective was to develop a model that could facilitate early intervention strategies. The proposed ANN architecture generated results in the form of four binary classifications based on student achievement: Pass - Fail, Distinction - Fail, Distinction - Pass and Dropout - Pass. The model achieved high classification accuracy, ranging from 84% to 94%, which surpassed the performance of the Logistic Regression models (79.82% - 85.60%) and SVM (79.95% - 89.14%).

Wilson chango et al. [14] addressed the challenge of student performance prediction by employing data fusion techniques on self-collected datasets. Four data fusion methods were investigated: merging all attributes, selecting the best attributes, combining multiple models (ensembles), and a combination of ensemble with feature selection. The authors evaluated the effectiveness of these methods using six classification algorithms: JRip, Nnge, PART, REPTree, Random Tree, and Avg trained on the fused datasets and Accuracy and AUC metrics were employed for model performance assessment. The results indicated that the overall method, which likely combines both ensemble and feature selection techniques, achieved the highest average accuracy 83.29%.

Al-Zawqari et al. [11] aimed to improve assessment of learner comprehension in online and blended learning environments. Their study proposed a novel approach for automatic feature extraction from raw data, eliminating the need for manual feature identification common in previous work. This method aimed to enhance model interpretability. To evaluate this approach, they constructed a random forest binary classifier for four student ability groups on the OULAD [6]. Additionally, they incorporated techniques to identify attributes most influential in predicting at-risk and dropout students. Furthermore, the authors developed an artificial neural network model to classify learner abilities, achieving high accuracy results (81% - 93%).

Vito Renò et al. [16] investigated the application of multimodal learning analytics in online learning environments for learner assessment. Their focus was on developing techniques that leverage the rich interaction data available in Open University Learning Analytics Datasets to predict student performance. The authors employed a supervised statistical machine learning model, specifically a Random Forest, to forecast a binary outcome (Pass/Fail). The trained model achieved an accuracy of 94.32%, indicating its potential for multimodal learner evaluation.

Al-Azazi et al. [17] contribute to the growing body of research on effective decision tree-based machine learning models by proposing a novel approach. They leverage Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures to predict student performance using frequently updated data. Their model treats OULAD as a time series dataset, allowing it to capture temporal dependencies. The results indicate that the proposed method achieves superior performance compared to a combination of individual students, with the Accuracy score reaching up to 95%. This study suggests the model's ability to learn generalizable features from the data. However, limitations exist, including potentially lengthy training times and complex installation procedures.

Similar to Al-Azazi et al. [17], Zhaoyu Shou et al. [10] proposed a Long Short-Term Memory and Artificial Neural Network based model for predicting learner outcomes over time. Their approach transforms OULAD into a multidimensional time series format and utilizes binary classification to identify students at high risk of failing. This allows for early intervention by instructors. The reported accuracy of this model ranges from 74% to 99.08%, with an F1 score following a similar trend from 73% to 99%.

3 Dataset

3.1 Description

This study employs the Open University Learning Analytics Dataset [18] to evaluate the proposed model. The dataset comprises aggregated clickstream data from students' interactions within the online learning environment, along with demographic information and assessment scores. This enables the construction of machine learning models to analyze student behavior based on their online platform activities.

OULAD comprises 22 classes from 7 distinct subjects: 4 courses in science, technology, engineering, and mathematics (STEM fields) and 3 courses in social sciences. Each course was offered numerous times throughout different semesters, with each unique code representation as "B" or "J" to identify the starting point in February or October within the years 2013 and 2014, respectively. The dataset is structured within ".csv" files interconnected through unique course identifiers. This structure facilitates the analysis of data encompassing 32,592 student identifications, 10,655,280 daily clickstream interactions and individual assessment scores. A detailed description of the database schema is as follows:

- courses.csv: Containing a list of all classes and their course codes.
- assessments.csv: Containing information about tests belonging to specific subjects.
- vle.csv: Stores information about existing learning materials in the online learning environment such as html pages, PDF files....
- studentInfo.csv: Containing demographic information and learning abilities of students.
- student Registration.csv: Containing information about the student's registered subjects.
- studentAssessment.csv: Containing student assessment results.
- studentVle.csv: Stores information that interacts with learning materials on the learner's online learning platform.

OULAD evaluates each student's learning abilities in four categories: excellent (Distinction), pass the course (Pass), fail the course (Fail) and drop the class (Withdrawn). As a result, our proposal concentrated on creating machine learning models to handle the multi-class classification and binary classification problems for this benchmark dataset.

3.2 Preprocessing

Given that OULAD is composed of individual comma-separated values files, merging these files is necessary to enable effective analysis. We employed the *pandas* library's *merge* function, utilizing *student_id* as the primary key table. Following this data integration process, the resulting dataset is structured into nine distinct CSV files. Each line in these files represents information pertaining to a single student. Two of these files encompass demographic information and total visit counts (clicks). The remaining seven files house assessment scores and corresponding labels, categorized by the seven offered subjects. A visual representation of the data processing steps is provided in Fig. 2.

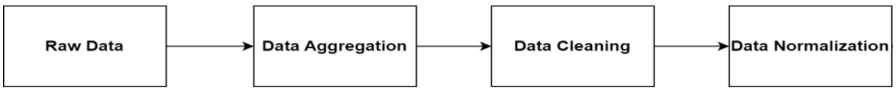

Fig. 2. Overall data preprocessing pipeline.

3.2.1 Demographics

The Open University collected demographic information on participants, including student ID, major, gender, birthplace, GPA, enrollment history (number of times registered for classes), and any declared special needs. As this data was initially provided in string format, we employed Label Encoding method [19] to transform it into a numerical representation suitable for machine learning model computations.

3.2.2 Assessment scores

Our analysis encountered missing assessment score data in a substantial number of courses. These missing values likely stem from students not submitting assignments. To address this, we imputed a value of zero to represent these absences. OULAD utilizes codes to represent 7 main subjects: AAA, BBB, CCC, DDD, EEE, FFF and GGG, each subject encompasses multiple classes categorized by year. Furthermore, each course details the number and type of lessons offered. Three distinct test types are coded within the data: TMA (Tutor Marked Assessment), CMA (Computer Marked Assessment) and Exam (Final Exam). To facilitate data merging, we proposed an approach utilizing average score data. This calculation incorporates test score columns and their corresponding weights, as defined in formulas (1a) and (1b).

The average score is a number between 0 and 100, and is calculated from the student's individual test scores after multiplying by the corresponding weights provided in the assessments.csv file. Let e be the type of test, n be the number of tests, the formula for calculating the average score is divided into 2 cases as follows:

- Calculate the total score of two types of tests when the participant receives either TMA or CMA (1a).

$$\sum_{i=1}^{n}(\mathrm{e}), e \in \{TMA, CMA\} \tag{1a}$$

- To incorporate all three values (TMA, CMA, and Exam) for an average score, add them all together and divide by 2 (1b).

$$\frac{1}{2}\sum_{i=1}^{n}(\mathrm{e}), e \in \{TMA, CMA, Exam\} \tag{1b}$$

We evaluate the performance of the chosen approaches across 8 datasets. These datasets comprised 7 individual ".csv" files, each corresponding to data from a single participant, and an additional, aggregate dataset that combined data from all participants.

3.2.3 Virtual learning interactions

Log files generated by the online learning platform Moodle are statistically recorded in the studentVLE.csv and vle.csv files. These logs capture the total number of each student's interactions (clicks) with various learning items. We normalize each feature within the three data sources to a range of [0, 1]. This normalization mitigates the impact of varying scales between different attribute fields. It's important to note that OULAD exhibits an imbalanced distribution of output classes. The class distribution is as follows: 42% Pass, 23% Fail, 10% Distinction, and 25% Withdraw, hence, addressing class imbalance is a crucial aspect of multi-class classification task. In this study, we employ the SMOTE oversampling technique [20] to balance the number of samples in each class of the training data. The distribution of OULAD before and after applying oversampling technique is shown in Fig. 3.

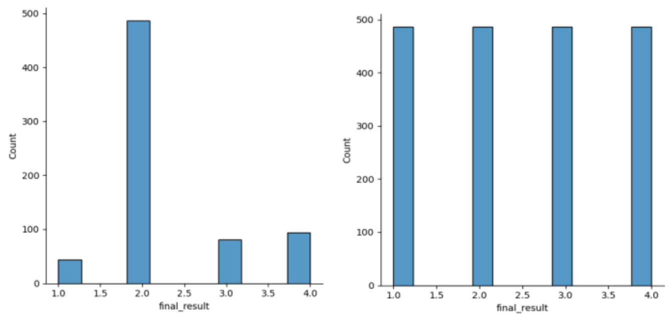

Fig. 3. Distribution of OULAD before and after applying an oversampling technique.

Furthermore, OULAD exhibits a high dimensionality, characterized by a large number of features. To address this challenge and enhance model efficiency, we employed a feature selection technique. This technique involved calculating the Pearson correlation coefficient (Eq. 2) between features to identify and remove redundant attributes with minimal impact on the model's output. Fig. 4a displays a heatmap of attribute correlations after the removal of less important features.

$$\sigma_{X,Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}. \tag{2}$$

Where: μ_X: is the mean of X.
μ_Y: is the mean of Y.
σ_X: is the standard deviation of X.
σ_Y: is the standard deviation of Y.

In the machine learning model development process, evaluating feature importance is vital for reducing training data size, enhancing model quality, and improving interpretability. Feature importance scores indicate a feature's impact on the model, with higher scores reflecting greater significance. We calculate feature importance using the *feature_importances* function from the *sklearn* library, which builds a RF model across the dataset and computes the Gini index for each feature. The Gini index (3), a measure introduced in the CART decision tree algorithm, assesses the purity of data at tree nodes, with higher values signifying greater feature importance. Fig. 4b highlights the most important features after the removal process.

$$Gini = 1 - \sum_{i=1}^{C} (p_i)^2 \tag{3}$$

Where: C is the number of class and $p_i = \frac{n_i}{N}$ with n_i is the number of samples belong to class *i*, N is the total number of samples

(a)

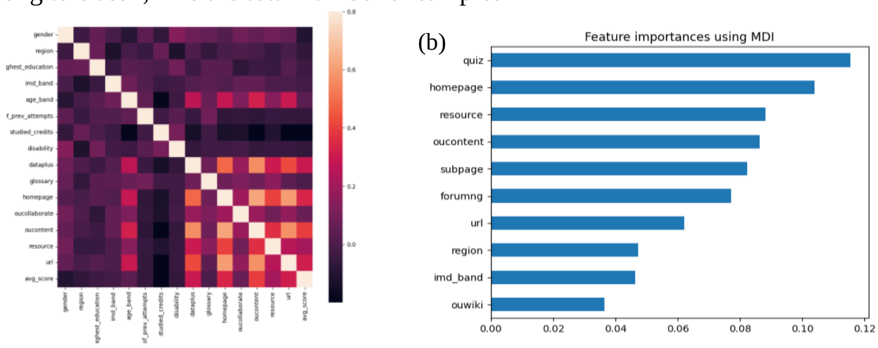

Fig. 4. (a) Heatmap represents the correlation of each attribute after removing unimportant features. (b) Top 10 most important features.

4 Methodology

This study proposes a multimodal learning analytics model utilizing different fusion techniques to predict student ability based on four categories: Distinction, Pass, Fail and Withdraw using data from diverse sources. We leverage OULAD, pre-processing it to extract three key data sources: demographic information, online platform interactions, and assessment scores. These data sources, originally residing in separate files, are merged into a unified dataset using Pandas' merge function with *student_id* as the primary key. The resulting dataset, following normalization detailed in Section 3, comprises 25,793 rows and 31 columns. This includes a single label column indicating student performance and a student identification column. We then calculate feature correlations and feature importance within the merged dataset to identify and remove irrelevant attributes. After data rebalancing, we employ three machine learning models -

Random Forest (RF), Support Vector Machine (SVM), and k-Nearest Neighbors (kNN) - to address the multi-class classification and binary classification problems. Finally, the model achieving the highest accuracy is subsequently selected. The implementation process is visually summarized in Fig. 5.

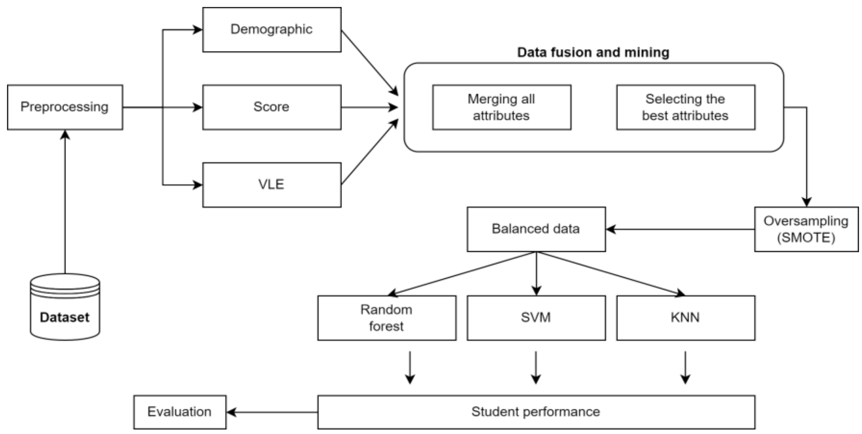

Fig. 5. The proposed student performance prediction model in multimodal learning analytics.

To determine the efficacy of our proposed model, we performed an experiment employing a late fusion multimodal learning analysis approach visualized in Fig. 6. This technique involves constructing separate machine learning models for each data source, followed by the integration of their outputs for the final prediction.

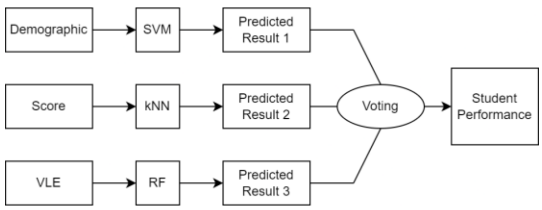

Fig. 6. Late fusion method with major voting approach for OULAD

The outputs of each model are merged using the major voting algorithm as follows: Each model assigns an integer label between 1 and 4 to each data sample. The final synthetic output is determined by the most frequent label predicted by RF, SVM and KNN models. In case all three models predict different labels, one of the three labels is chosen randomly as the final output.

5 Experiment results

We utilize accuracy and F1-score to gauge the model's performance. The formula for calculating accuracy is presented in Equation (4).

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (4)$$

The F1-score serves as a harmonic mean, balancing precision and recall to evaluate the model's effectiveness in correctly identifying positive class instances. The formula for calculating accuracy is presented in Equation (5).

$$F1\ score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (5)$$

5.1 Result of the proposed model

This section presents experimental results of proposed models on OULAD using k-fold cross-validation (k=5). Table 1 compares the accuracy of three machine learning models - RF, SVM and kNN - both before and after applying oversampling techniques across seven distinct datasets: AAA, BBB, CCC, DDD, EEE, FFF and GGG, as well as a combined dataset. Before oversampling, RF's accuracy ranged from 66.43% to 82.41%, with an overall average of 71.46%. After oversampling, RF's accuracy improved to a range of 80.66% to 93.15%, with an overall average of 82.31%. SVM's accuracy before oversampling ranged from 66.19% to 78.43%, improving post-oversampling for most datasets, though some decreased. KNN showed accuracy improvements across all datasets after oversampling, from 63.72% to 75.41% overall. Among the three machine learning models, the RF model exhibited the most significant improvement after applying the oversampling technique. The kNN model also demonstrated a notable improvement, with the accuracy of the entire dataset increasing from 64.72% to 75.41%. Although the SVM model showed improved results across most datasets, the accuracy decreased specifically for the CCC and GGG datasets due to the particular characteristics of these datasets when the model was applied.

Table 1. Accuracy results for 8 datasets. The table compares the performance of the models before and after applying oversampling techniques and removing unimportant attributes.

Model	**Over-sampling**	**AAA**	**BBB**	**CCC**	**DDD**	**EEE**	**FFF**	**GGG**	**Total**
RF	Before	82.41	68.57	74.72	74.49	68.77	72.75	66.43	71.46
	After	**93.15**	**83.13**	**80.66**	**85.58**	**86.44**	**85.34**	**80.66**	**82.31**
SVM	Before	78.43	67.36	73.40	73.80	67.38	69.40	66.19	69.38
	After	81.23	68.74	68.36	76.45	70.69	73.59	55.21	68.80
kNN	Before	75.46	64.00	65.14	68.08	63.85	63.89	55.43	63.72
	After	84.52	76.15	69.47	77.85	79.52	75.75	70.44	75.41

Table 2 presents a comparative analysis of the F1 scores of the RF, SVM, and kNN models before and after the application of oversampling, across seven individual datasets and a combined dataset. Initially, the RF model achieved an overall F1 score of 63.84% for the entire dataset, with individual dataset scores ranging from 39.87% (GGG) to 78.16% (AAA). Following oversampling, the overall F1 score improved to 82.08%, with individual scores ranging from 80.71% to 93.14%. Notably, the GGG dataset exhibited the most significant improvement, with the RF model's score increasing by 40.84%, underscoring the efficacy of oversampling in enhancing RF model performance. For the SVM model, the initial F1 score was 59.28%, with a range from

35.93% (GGG) to 65.55% (CCC). Post-oversampling, the F1 score improved to 67.56%, with scores ranging from 53.96% (GGG) to 80.65% (AAA). Although the improvement was less pronounced compared to the RF model, it was still noteworthy, particularly for the AAA dataset. The kNN model initially recorded an F1 score of 53.70%, with a range from 38.54% (GGG) to 58.54% (CCC). After oversampling, the kNN model's F1 score markedly increased to 74.62%, with scores ranging from 69.04% to 83.72%, demonstrating significant improvement, especially in the AAA dataset.

Table 2. F1-scores achieved by our model on 8 datasets. The table compares the performance before and after applying two data pre-processing techniques: oversampling and attribute selection.

Model	**Over-sampling**	**AAA**	**BBB**	**CCC**	**DDD**	**EEE**	**FFF**	**GGG**	**Total**
RF	Before	78.16	55.97	66.74	67.17	53.91	66.75	39.87	63.84
	After	**93.14**	**82.84**	**80.44**	**85.38**	**86.26**	**85.10**	**80.71**	**82.08**
SVM	Before	42.26	47.67	65.55	62.50	43.52	60.45	35.93	59.28
	After	80.65	66.77	66.62	75.54	70.09	72.61	53.96	67.56
kNN	Before	44.18	51.18	58.54	55.58	48.73	53.40	38.54	53.70
	After	83.72	75.34	69.22	77.39	78.14	74.86	69.04	74.62

The results outlined in Tables 1 and 2 demonstrate that the RF model outperforms the other models evaluated, achieving the highest level of accuracy. These findings emphasize the importance of employing oversampling techniques when addressing imbalanced datasets. OULAD is predominantly classified under the Pass category, resulting in training bias and diminished effectiveness in predicting minority classes. Additionally, the removal of non-essential attributes from the dataset contributes to enhanced model performance. Therefore, the application of oversampling techniques combined with attribute selection can markedly improve the performance of machine learning models in real-world scenarios.

5.2 Results comparison

- Between different data fusion methods

Table 3. Comparison between early fusion method and late fusion method

Module	**Sample size**	**Early fusion Accuracy**	**Late fusion Accuracy**	**Early fusion F1 score**	**Late fusion F1 score**
AAA	704	**93.15**	78.41	**93.14**	78.55
BBB	6,054	**83.13**	65.93	**82.84**	65.35
CCC	3,410	**80.66**	64.94	**80.44**	64.15
DDD	4,932	**85.58**	67.68	**85.38**	67.46
EEE	2,296	**86.44**	68.60	**86.26**	68.61
FFF	6,287	**85.34**	71.76	**85.10**	71.74
GGG	2,109	**80.66**	52.74	**80.71**	51.00
Total	25,793	**82.31**	65.74	**82.08**	64.94

Our proposed early fusion model achieved superior performance on 8 datasets (AAA, BBB, CCC, DDD, EEE, FFF, GGG, Total) compared to the late fusion method. These results were shown in Table 3 with accuracy, and the F1 score metrics respectively.

- With previous studies

The efficacy of our proposed model was further evaluated through a comparative analysis with previous studies utilizing similar data sources - specifically, demographic information, assessment scores, and virtual learning interactions - to tackle the multi-class classification and binary classification tasks of predicting student performance. Al-Azazi et al. [17] utilized the OULAD dataset with an ANN-LSTM model for four output classes: Distinction, Passed, Failed and Withdraw, achieving 72% accuracy and an F1-score of 66%. Similarly, Shou [10] introduced a model combining a multi-layer LSTM with a multi-head self-attention mechanism, followed by a neural network, reaching 74% accuracy and a 73% F1-score. However, both approaches have limitations, including a lack of consideration for data imbalance and the computational complexity of deep learning models. Our method addresses these issues by incorporating data rebalancing before model training and using feature importance analysis to remove irrelevant attributes, thereby enhancing efficiency. As shown in Table 4, our RF model achieves 82.31% accuracy and an F1-score of 82.08%, significantly outperforming the ANN-LSTM and MTAPSP models by 10.31% and 8.31% respectively.

Table 4. Comparison results between proposed model and ANN-LSTM, MTAPSP in multi-class classification setting.

Evaluation Metric	ANN-LSTM Fatima Ahmed Al-azazi et al. (2023)	MTAPSP Zhaoyu Shou et al. (2024)	RF Our model
Accuracy	72	74	**82.31**
F1 Score	66	73	**82.08**

To further assess our approach, we also compare it with previous studies utilizing binary classification models. Waheed et al. [6] reduced the feature set from 54 to 30 before training an ANN with three hidden layers. Al-Zawqari [11] developed four independent binary classifiers for Pass-Fail, Distinction-Fail, Distinction-Pass, and Withdraw-Pass using OULAD, producing predictions across four quarters (Q1-Q4). Building on these studies, we created four separate binary classifiers using 29 features from the entire dataset, as outlined in Section 3. We employed three machine learning models - SVM, kNN and RF - along with the data processing techniques described in Section 3.

Table 5 shows that our model achieves competitive accuracy scores, demonstrating its effectiveness. Our proposed model attained a peak classification accuracy of 97.9% for distinguishing between "Withdraw" and "Pass" classes, surpassing Waheed's ANN method by 6.3%. The lowest accuracy observed in our study was 89.46% for classifying "Distinction" and "Pass" classes, which still exceeds the accuracy achieved by both Waheed's ANN method and Ali Al-Zawqari's model. In all four binary classification tasks evaluated, our model consistently outperforms previous studies.

Table 5. Comparison results between proposed model and Simulated ANN and RF in binary classification setting.

Binary classification	Simulated ANN Waheed et al. (2020)	RF Ali Al-Zawqari el al. (2022)	kNN Our model	SVM Our model	RF Our model
Pass - Fail	85.75	85.83	86.67	86.51	**91.28**
Distinction - Fail	88.95	87.29	93.34	94.43	**96.91**
Distinction - Pass	81.21	81.26	80.72	81.10	**89.46**
Withdraw - Pass	93.28	92.90	96.36	96.62	**97.90**

6 Conclusion

This study investigated the development of a student performance prediction model within the context of multimodal learning analytics. Leveraging the publicly available dataset called OULAD, the model incorporates early fusion techniques to integrate data modals from different sources. Our experiments demonstrate that the adopted Random Forest model achieves competitive accuracy compared to previous works. These findings highlight the potential of the student performance prediction model in multimodal learning analytics as a powerful tool for comprehensive learner assessment. As a result, such a model can help guide teachers to targeted interventions and modifications to improve the learning experience for individual students. While the current model demonstrates effectiveness when trained on OULAD, it inherently possesses limitations due to its dependence on retrospective student data for performance prediction. To maximize the model's utility, early intervention through learner capability forecasting is crucial. This would empower students with self-awareness and facilitate the development of appropriate learning strategies within the educational setting.

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